

4 (long)

- (a) Find the point of intersection P of the line $\underline{r} = (1,3,0) + \lambda (1,-1,1)$ and the plane $z = 2y$. [6]
- (b) The volume of a tetrahedron is given by $\frac{1}{3} \times \text{base area} \times \text{height}$. Show that the volume can be written as $\frac{1}{6}$ of the magnitude of the scalar triple product of the vectors representing the edges of the tetrahedron that meet at a vertex. [10]
- (c) Find the volume of the tetrahedron whose vertices are: the origin, the point P found in part (a), the point $(1,3,0)$ and the point $(\alpha,0,0)$. [4]
- (d) Find the shortest distance from the point $(1,3,0)$ to the opposite face of the tetrahedron defined in part (c). [10]

5 (long) The differential equation governing the output voltage V of a circuit subject to a switching input is

$$\frac{d^2V}{dt^2} + 4V = \begin{cases} 1 & 0 < t \leq \pi/2 \\ 0 & t > \pi/2 \end{cases}$$

where the voltage V and dV/dt are continuous at the point of switching and where $V = 0$ and $dV/dt = 0$ at $t = 0$.

- (a) Find V for t in the range $0 < t < \pi/2$. [5]
- (b) Find V for all $t \geq \pi/2$. [6]
- (c) If the right hand side of the differential equation is changed to 0.1 for $t > \pi/2$, find and sketch $V(t)$. [19]